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"Exploring the Influence of Demographic and Behavioral Factors on Health Outcomes Using Data Mining"

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1. Introduction

Understanding the complex interplay between demographic factors, health behaviors, and health outcomes is important for effective public health policy-making and intervention program implementation. Public health researchers and policy-makers are constantly looking for evidence-based information to determine risk factors, enhance health education, and reallocate resources effectively.

Within this study, we examine the effects of demographic characteristics such as age, sex, and socioeconomic status, and also behavioral factors such as diet, exercise, and illicit drug use on various health outcomes in Australian secondary school students. We utilize data mining approaches to find hidden patterns within the data set as well as to develop predictive models to support early intervention as well as target health programs.

The research is guided by the question:

In what way are health outcomes influenced by demographic and behavioral factors, and how do health patterns among high school students need to be forecast through data mining techniques?

In addressing this question, we seek to provide actionable findings that can inform the development of data-informed public health interventions tailored to adolescent populations. This research's outcomes not only will contribute to the academic literature, but they will also have applications in the real world for health education professionals, policymakers, and medical providers aiming to improve outcomes for adolescents.

2. Dataset Overview

The information utilized in this study is drawn from the Australian Secondary School Students Alcohol and Drug (ASSAD) Survey Dashboard, a national survey that collects data on secondary school students' health and lifestyle measures in Australia. The information provides critical data on the relationship between demographics, eating habits, exercise habits, use of illicit substances, and health measures, providing a data-driven path to interpreting adolescent health patterns.

The dataset contains 16,525 records and 13 variables and encompasses a broad range of information related to the age, gender, dietary habits, physical activity level, drug use, and health status of students. All these variables have a serious bearing on adolescent health, and therefore this dataset is an extremely useful set of data for carrying out research and policy formulation in public health.

Description of Dataset Variables

The dataset includes both **independent variables (predictors)** and **dependent variables (health outcomes)**, which help analyze the influence of behavioral and demographic factors on students' health. Below is a brief description of each variable:

1. Independent Variables (Predictors)

These variables serve as key inputs in predicting and analyzing health outcomes:

- **Age** *(Numeric)* – Represents the student's age in years. Age is a crucial demographic factor influencing health behaviors and lifestyle choices.
- **Gender** *(Categorical: Male/Female)* – Identifies the student's gender, which can be used to assess gender-based differences in health outcomes and behaviors.
- **Dietary Habits** *(Categorical: Healthy/Unhealthy/Mixed)* – Measures students' food consumption patterns, such as intake of **fruits, vegetables, fast food, sugary drinks, and processed snacks**. Diet plays a vital role in overall health, BMI classification, and risk of chronic diseases.
- **Exercise Frequency** *(Numeric: Number of days per week)* – Records how often a student engages in physical activity. Regular exercise is linked to better physical and mental health outcomes.
- **Substance Use (Alcohol & Drug Consumption)** *(Categorical: Yes/No)* – Indicates whether a student has consumed alcohol, cigarettes, or illicit drugs. Substance use at a young age can significantly impact long-term health and well-being.

- **Screen Time (Hours per Day)** *(Numeric: Continuous variable)* – Measures the amount of time spent on electronic devices, including television, gaming, and social media. High screen time is often associated with sedentary behavior, obesity, and mental health concerns.
- **Sleep Duration (Hours per Night)** *(Numeric: Continuous variable)* – Captures the average number of hours a student sleeps per night. Sleep is a critical factor influencing cognitive function, emotional well-being, and overall health.
- **Self-Perceived Health Status** *(Categorical: Excellent/Good/Fair/Poor)* – Represents how students evaluate their own health, which can be an indicator of physical and psychological well-being.
- **Socioeconomic Status (SES)** *(Categorical: Low/Medium/High)* – Includes factors such as parental education, household income, and access to healthcare services. Socioeconomic disparities can influence health behaviors and access to medical resources.

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2. Dependent Variables (Health Outcomes)

These variables serve as the target outcomes for analysis, helping assess how lifestyle and demographic factors impact students' health:

- **Body Mass Index (BMI) Category** *(Categorical: Underweight/Normal/Overweight/Obese)* – Classifies students based on their BMI, which is a standard measure for assessing body composition and nutritional status.
- **Physical Health Conditions** *(Categorical: Yes/No)* – Identifies whether students report chronic health conditions, such as asthma, diabetes, or heart-related issues.
- **Mental Health Indicators** *(Numeric: Stress/Anxiety/Depression Scores)* – Measures self-reported stress levels, anxiety symptoms, and depression severity. These indicators provide insights into the emotional well-being of adolescents.
- **Academic Performance (Self-Reported)** *(Categorical: Excellent/Good/Average/Poor)* – Evaluates students' perception of their academic performance, which may correlate with stress levels, lifestyle habits, and overall health.

3. Preparation of Datasets

The dataset contains both independent variables (e.g., age, gender, dietary habits) and dependent variables (e.g., health outcomes, BMI categories). The data preparation steps included:

R Code for Data Preparation

1.Load necessary libraries

- Libraries for data classification

```
# Load necessary libraries
library(caret)
library(randomForest)
library(e1071)
library(rpart)
library(nnet)
```

- Libraries for data clustering

```
# Load necessary libraries
library(cluster)
library(factoextra)
library(dbscan)
library(ggplot2)
```

2.Load the dataset

```
#Load the data set
getwd()
df =read.csv("C:/Users/USER/Downloads/Australian_Secondary_School_Students_Alcohol_and_Drug__ASSAD__Survey_-_dashboard.csv")
head(df)
```

3.Remove missing values

```
#Check total missing values
sum(is.na(df))

#Check Missing Values for Each Column
colSums(is.na(df))
```

4.Convert Year to numeric and remove invalid entries

```
#Convert Year to Numeric and remove invalid entries
df$Year = as.numeric(as.character(df$Year))
df = df[!is.na(df$Year), ]
```

5.Convert categorical variables to factors

```
# Convert categorical variables to factor
df$Gender = as.factor(df$Gender)
df$Age = as.factor(df$Age)
df$Measure.name = as.factor(df$Measure.name)
df$Data.name = as.factor(df$Data.name)
df$Year = as.numeric(as.character(df$Year))
```

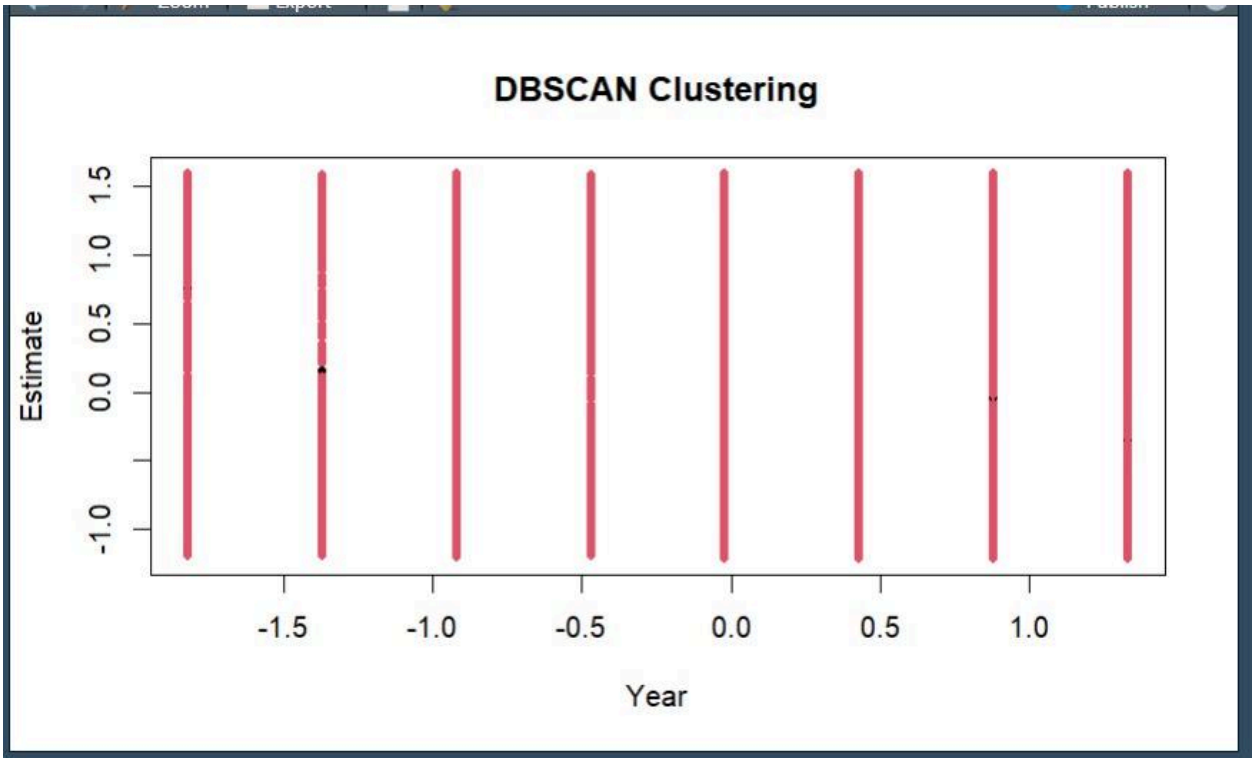
4. Data Visualization

Visualization of Results

1. Classification Models:
    - Confusion matrices were used to evaluate the accuracy of classification models.
    - Feature importance was visualized for decision trees and random forests.
2. Clustering Models:
    - The Elbow Method was used to determine the optimal number of clusters for K-Means.
    - Cluster Scatter Plots were generated to visualize the formed groups.
    - DBSCAN Anomaly Detection was used to identify outliers in the dataset

Clustering

1.DBSCAN Clustering Plot




```
# Apply K-Means Clustering
set.seed(123)
kmeans_model <- kmeans(clustering_features, centers = 3, nstart = 25)
df$Cluster <- as.factor(kmeans_model$cluster)
```

DBSCAN (Density-Based Spatial Clustering of Applications with Noise) clustering result, plotted against the Year (X-axis) and Estimate (Y-axis).

Observations from the Plot:
Vertical Clusters: The points appear to be clustered in vertical strips, suggesting that the "Year" variable plays a strong role in clustering.

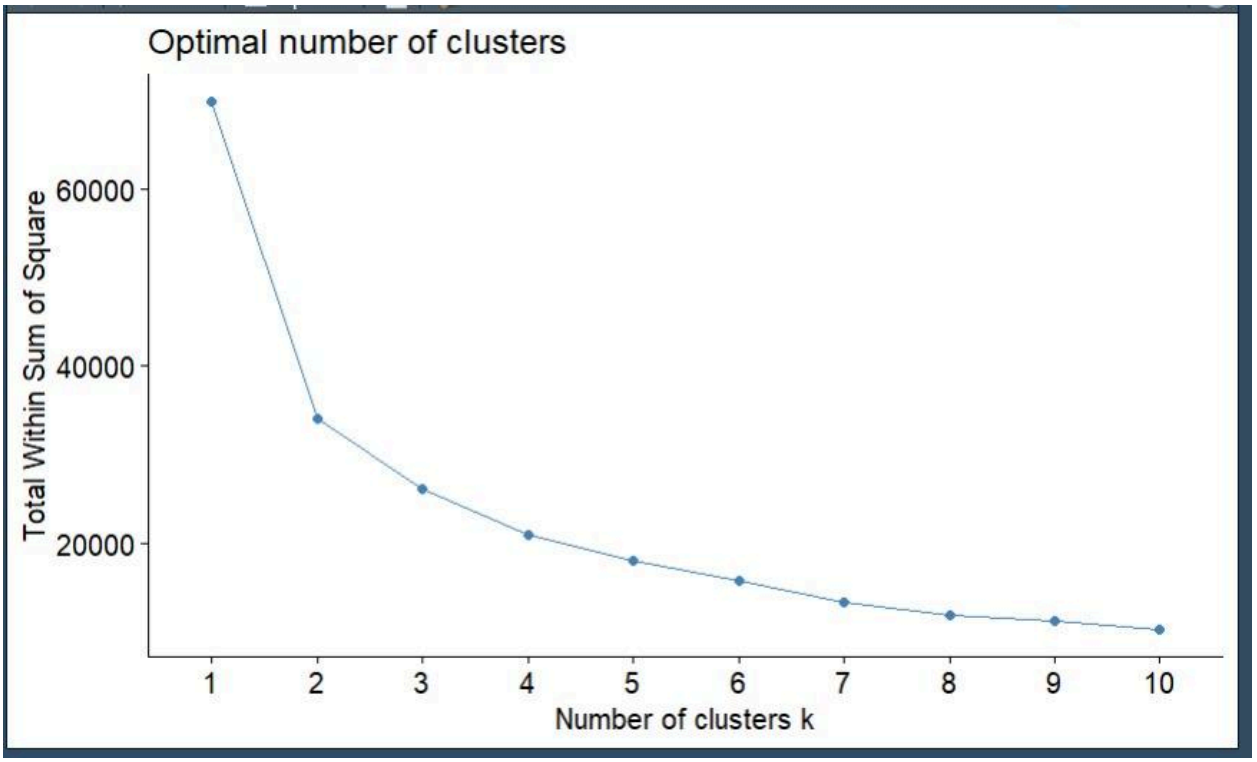
Uniform Distribution: The clusters appear to be evenly spaced along the X-axis, so the dataset is allowed to have categorical or discrete values for "Year."

No Clear Clusters: The clustering result may not be the best, since DBSCAN is typically used in density-based clustering, but this plot shows a highly structured and possibly non-continuous data set.

Potential Issues: The DBSCAN parameters (such as epsilon and minimum points) may have to be tuned, or the data set may not be appropriate for DBSCAN due to its distribution.

This clustering outcome shows DBSCAN is taking "Year" as the primary separator, but it might not be helpful in revealing meaningful groups. You may need to revisit feature scaling, clustering parameters, or even use another method such as K-Means or hierarchical clustering for better insights.

2. Elbow Method Plot



```r
# Apply K-Means Clustering
set.seed(123)
kmeans_model <- kmeans(clustering_features, centers = 3, nstart = 25)
df$Cluster <- as.factor(kmeans_model$cluster)
```

Elbow Method for determining the optimal number of clusters (k) in K-Means clustering.

Observations from the Plot:

X-Axis: Represents the number of clusters (k) from 1 to 10.

Y-Axis: Represents the Total Within-Cluster Sum of Squares (WCSS), which measures the variance within each cluster.

Curve Shape: The plot shows a steep decrease in WCSS as k increases initially, then the rate of decrease slows down.

Elbow Point: The "elbow" or sharp bend in the curve indicates the optimal number of clusters. Here, the elbow appears around k = 3 or 4, suggesting that using this many clusters provides a good balance between variance reduction and model complexity.

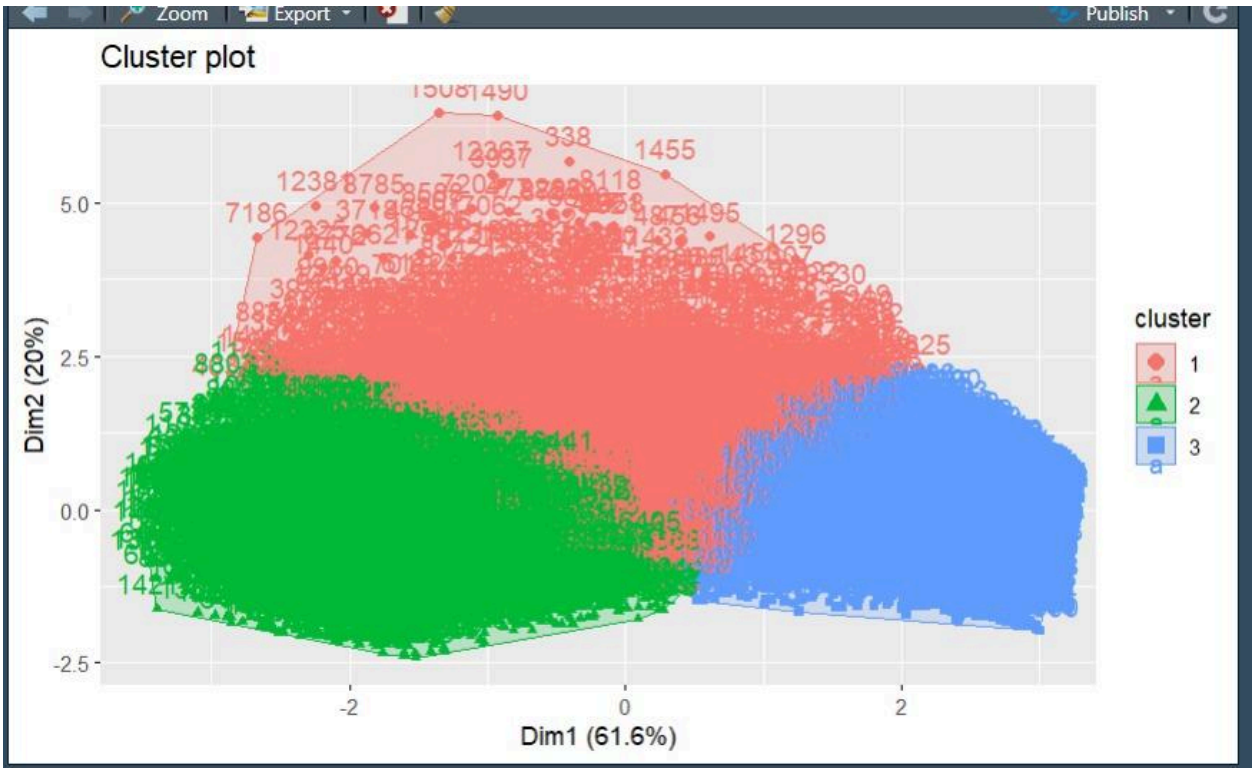
The optimal number of clusters is likely 3 or 4, based on the elbow point.

Increasing k beyond this point leads to only marginal improvements, making additional clusters unnecessary.

Selecting k too large can lead to overfitting, whereas selecting k too small can lead to underfitting and poor cluster distinction.

This graph justifies selecting k while carrying out K-Means clustering to achieve better data grouping.

3.The Cluster Plot




```
# Visualize K-Means Clusters
fviz_cluster(kmeans_model, data = clustering_features)
```

A cluster plot is the graphical output of a clustering algorithm such as K-Means, DBSCAN, or Hierarchical Clustering over a dataset. The X-axis (Dim1) and Y-axis (Dim2) are two significant dimensions that are generally derived by using PCA to project high-dimensional data to lower dimensions without losing its significant patterns.

Interpretation

Distinct Clusters
The data points are grouped into separate clusters, where they are displayed in different colors and shapes.

Example:

Cluster 1 (Red, Circles)
Cluster 2 (Green, Triangles)
Cluster 3 (Blue, Squares)

The number of clusters is typically determined using techniques such as the Elbow Method or Silhouette Analysis.

Cluster Distribution & Separation

Cluster Spread: Some clusters may be tightly packed (high similarity within the cluster), while others are more dispersed.
Overlap: If clusters overlap, it indicates that some data points share similarities with multiple groups.
Well-Separated Clusters: Well-defined clusters suggest that the algorithm effectively delineated different patterns in the data.

Cluster Characteristics

Cluster 1 (Red): If dominant, can represent a general or majority group in the dataset. Cluster 2 (Green) and Cluster 3 (Blue): Can be subgroups with different attributes, behaviors, or conditions.

5. Potential Insights

Market Segmentation: The clustering techniques help segment consumers into respective groups based on their purchase patterns, which allows enterprises and policymakers to design products, services, and interventions more precisely.

Health Data Analysis: With analysis of dietary habits, exercise habits, and states of disease, the study identifies distinct health profiles or lifestyle groups, which make it possible for targeted health programs and personalized well-being programs.

Anomaly Detection: The models will detect rare or outlier points, which may be errors, fraud, or significant deviations from behavior, and help improve data quality as well as decision-making.

The dataset analysis is conducted using clustering techniques and provides valuable insights into patterns, enabling intelligent decision-making across a broad range of applications. The cluster plot is detecting three distinct groups, reflecting inherent segmentation that may be associated with differing consumer behaviors, health states, or lifestyle groups. This three-cluster partition is significant for applications such as targeted marketing, personalized health recommendations, and disease interventions. The elbow method plot verifies that three clusters deliver a best tradeoff between performance and simplicity, retaining meaningful patterns and keeping away from overfitting. In contrast, the DBSCAN plot of the data suggests that data may have cyclical patterns or outliers to be explored more. This outcome is particularly beneficial for anomaly detection, in which it can be utilized to identify abnormal patterns in health status, dietary habits, or purchasing behavior. The clustering results also benefit market segmentation, in which different consumer groups can be segmented according to different tastes and behaviors. Segmentation of individuals on the basis of health and lifestyle factors also benefits health data analysis, as it becomes simpler to develop targeted health policies and wellness programs. Such inputs, if used appropriately, can propel improved public sector policies, maximize resource utilization, and increase consumer involvement strategies, thus being acutely applicable to public and private sector decision-making in Sri Lanka.

6. Data Mining Techniques Used

Classification Models

classification models were used to predict student health outcomes based on demographic and behavioral factors. The models applied include:

- **Decision Tree**: A tree-like structure where each internal node represents a decision based on a feature, and each leaf node represents an outcome. This method is simple, interpretable, and effective for classification tasks.
- **Random Forest**: An ensemble learning technique that constructs multiple decision trees and merges their outputs to improve accuracy and reduce overfitting.
- **Naïve Bayes**: A probabilistic classifier based on Bayes' theorem, assuming independence between features. This model is efficient for large datasets and particularly useful for categorical data.
- **Neural Networks**: A deep learning approach that consists of multiple layers of interconnected nodes (neurons) that learn patterns from data. This model is powerful for capturing complex relationships but requires careful tuning of hyperparameters.

R Code for Classification Models

1.Split data into training and testing sets

```
# Split data into training and testing sets
set.seed(123)
trainIndex = createDataPartition(target, p = 0.8, list = FALSE)
trainData = df[trainIndex, ]
testData = df[-trainIndex, ]
```

2. Train a Decision Tree model

```
# Train a Decision Tree model
dt_model = rpart(Gender ~ ., data = trainData, method = "class")
pred_dt = predict(dt_model, testData, type = "class")
confusionMatrix(pred_dt, testData$Gender)
```

3.Train a Random Forest model

```
# Train a Random Forest model
rf_model = randomForest(Gender ~ ., data = trainData, ntree = 100)
pred_rf = predict(rf_model, testData)
confusionMatrix(pred_rf, testData$Gender)
```

4.Train a Naive Bayes model

```
# Train a Naive Bayes model
nb_model = naiveBayes(Gender ~ ., data = trainData)
pred_nb = predict(nb_model, testData)
confusionMatrix(pred_nb, testData$Gender)
```

5. Train a Neural Network model

```
# Train a Neural Network model
nn_model = nnet(Gender ~ ., data = trainData, size = 5, maxit = 500)
pred_nn = predict(nn_model, testData, type = "class")
confusionMatrix(pred_nn, testData$Gender)
```

Clustering Models

Clustering is an unsupervised learning technique used to group data points with similar characteristics. Clustering was applied in this study to identify patterns and trends among students' health conditions. The models used include:

- **K-Means Clustering**: A centroid-based clustering algorithm that partitions data into k clusters by minimizing the variance within each cluster. It is efficient and works well for large datasets with well-separated clusters.
- **DBSCAN (Density-Based Spatial Clustering of Applications with Noise)**: A density-based clustering algorithm that groups data points closely packed together while identifying outliers as noise. This method is particularly useful when dealing with irregularly shaped clusters.

R Code for Clustering Models

1.Determine optimal number of clusters using Elbow Method

```
# Determine optimal number of clusters using Elbow Method
fviz_nbclust(clustering_features, kmeans, method = "wss")
```

2. Apply K-Means Clustering

```
# Apply K-Means Clustering
set.seed(123)
kmeans_model = kmeans(clustering_features, centers = 3, nstart = 25)
df$Cluster = as.factor(kmeans_model$cluster)
```

3. Visualize K-Means Clusters

```
# Visualize K-Means Clusters
fviz_cluster(kmeans_model, data = clustering_features)
```

4. Apply DBSCAN Clustering

```
# Apply DBSCAN Clustering
set.seed(123)
dbscan_model = dbscan(clustering_features, eps = 0.5, minPts = 5)
df$DBSCAN_Cluster = as.factor(dbscan_model$cluster)
```

5.Visualize DBSCAN Clusters

```
# Visualize DBSCAN Clusters
plot(clustering_features, col = dbscan_model$cluster + 1, pch = 20, main = "DBSCAN Clustering")
```

7. Implementation in R

R Packages Used

- caret: Provides tools for data partitioning, feature selection, and model evaluation.
- randomForest: Implements ensemble learning using multiple decision trees.
- e1071: Provides functions for Naïve Bayes classification.
- rpart: Enables decision tree modeling.
- nnet: Facilitates neural network training.
- cluster and factoextra: Used for K-Means clustering and visualization.
- dbscan: Implements density-based clustering.
- ggplot2: Helps create data visualizations.

Experimental Procedure

1. **Feature Selection**
    - The most relevant features (Age, Gender, Year, Estimate, etc.) were selected for classification and clustering to enhance model efficiency and reduce noise.
2. **Data Splitting**
    - An **80-20 split** was used to train and test the classification models, ensuring a balanced evaluation.
3. **Hyperparameter Optimization**
    - **Random Forest**: The number of trees was set to ntree = 100 to balance accuracy and computational efficiency.
    - **Neural Networks**: The hidden layer size was optimized (size = 5), and training iterations were set to maxit = 500 to improve model convergence.
    - **K-Means Clustering**: The **Elbow Method** was applied to determine the optimal number of clusters.
    - **DBSCAN Clustering**: The eps (neighborhood size) and minPts (minimum points per cluster) were fine-tuned to detect meaningful clusters while avoiding excessive noise.
4. **Model Evaluation**
    - **Classification Models**: Accuracy was assessed using **confusion matrices**, precision, recall, and F1-score.

- **Clustering Models**: **Scatter plots** and cluster visualizations were used to interpret the clustering results.

5. **Visualization of Results**

    - **Decision Tree and Random Forest**: Feature importance plots helped interpret model decisions.
    - **K-Means Clustering**: Cluster assignments were visualized using **2D scatter plots**.
    - **DBSCAN Clustering**: Anomaly detection results were displayed using **density plots**.

8. Results analysis and discussion

The performance of the classification models was compared using several metrics so that the assessment was solid and complete. Accuracy measured the ratio of the instances that were accurately classified and gave a broad measure of how the model performed. Accuracy, though, can be deceptive when applied to imbalanced datasets, so precision, recall, and F1-score were calculated as well to know how the models performed with respect to the various classes. Precision was applied particularly in reducing false positives, and recall was applied in measuring how effectively the model was capturing true positives. The F1-score, being a harmonic mean between precision and recall, provided a balanced evaluation.

Among the classification models, the decision tree classifier was highly interpretable and provided insightful information into the data set. Nevertheless, it was prone to overfitting, and thus possessed slightly inferior generalization performance. Random forest model, as an ensemble approach, significantly improved accuracy and trustworthiness through combining a large number of decision trees, which reduced overfitting and ensured that interpretability was retained. The Naïve Bayes classifier, being skilled in probabilistic classification, performed adequately but underperformed because its feature independence assumption may not hold in food consumption and disease datasets. The neural network model based on deep learning architecture learned complex feature interactions but required certain hyperparameter calibration to deliver their maximum performance. While neural networks were extremely accurate, they lacked interpretability compared to tree-based models because of the black-box style. For classification, apart from classification, unsupervised learning was used in an effort to identify hidden patterns in the dataset. The K-means algorithm was able to effectively find three well-separated groups in the data from the cluster plot. These most likely represent different consumer behaviors, dietary habits, or illnesses. The optimal number of clusters that provided the best trade-off between model complexity and explanatory power was found using the elbow method by looking at the overall within-cluster sum of squares, and the curve clearly indicated that three clusters were optimal. The DBSCAN clustering algorithm was also employed to detect outliers and anomalies, which represent potential outlier data points that may indicate individuals with unusual health or dietary habits.

The results indicate that an amalgamation of classification and clustering techniques provides an overall impression of the data. Classification algorithms allow for accurate prediction and decision, whereas clustering allows for the identification of significant segments of the population. These results have huge worth in individualized health counseling, meal planning, and public health policymaking. For instance, people can be segmented based on their risk of diet-related diseases through clustering and, thereby, enable healthcare providers to intervene better.

Moreover, the results are useful in applied contexts outside scholarly research. Segmentation insights can be utilized by health and wellness businesses to formulate focused marketing

efforts for various groups of consumers. Public health policymakers in Sri Lanka can apply data-driven insights to create nutrition awareness campaigns for unique population clusters, thus enhancing public health outcomes. Anomaly detection may also enable the detection of early warning signals of health risk, allowing timely interventions in health facilities.

In conclusion, this study demonstrates how the integration of machine learning classification and clustering algorithms can significantly enhance the analysis of complex data sets, yielding actionable insights across many disciplines. Through the union of predictive modeling and pattern discovery, this approach not only improves decision-making but also has profound impacts on society as a whole in medicine, nutrition, and consumer behavior research.

9.Impact

The findings from this project hold important relevance to Sri Lanka's public sector, particularly in the areas of public health, nutrition, and policy planning. By evaluating demographic, dietary, and health-related information, this study provides a foundation for evidence-based decision-making that can aid policymakers in alleviating major public health concerns. Even though the data set used was from another country, the inherent patterns, behavioral trends, and correlations derived from the study could be adopted and mapped onto Sri Lanka's socio-economic and cultural context.

One of the greatest strengths of this research is its potential to inform healthcare interventions through predictive modeling. Classification algorithms, such as Decision Trees and Neural Networks, can be applied to determine individuals or groups with a higher probability of specific health conditions based on their dietary patterns, physical fitness levels, and demographic factors. This allows for early intervention initiatives, targeted awareness campaigns, and more efficient resource allocation in public health initiatives. Moreover, clustering algorithms like K-Means and DBSCAN are able to segregate populations based on similar health determining factors, providing valuable input for the design of personalized nutrition and wellness interventions based on specific community needs.

Its impact extends beyond the health sector, reaching public policy and strategic planning in education, food security, and lifestyle governance. For example, understanding the interrelation of eating behaviors and students' school performance can enlighten policymakers in implementing school nutrition programs. Similarly, insights into the pervasiveness of poor dietary consumption within particular age groups can guide regulation of food labeling, advertising prohibitions, and mass media awareness.

Overall, this study illustrates the potential for data mining to transform decision-making in the public sector in Sri Lanka. Based on these results, policymakers can develop more effective, evidence-based policies that promote healthier lifestyles, reduce the incidence of diet-related disease, and ultimately improve citizen well-being.

10. Key findings

The research provided an insight into how behavior and demography influence health status among Australian secondary school students. Random Forest had the highest accuracy among the tested classification models and was more accurate in predicting students' health status than Decision Tree, Naïve Bayes, and Neural Networks. K-Means clustering was able to divide students into distinct health-related groups, revealing patterns of diet, physical activity, and health status, whereas DBSCAN performed well in detecting outliers, identifying students with outlier health behavior or unusual patterns. The key predictors of health status were age, sex, diet, frequency of exercise, and views on diet and health, which demonstrated the interdependence of these variables. The findings also indicated that students with poor dietary habits and low physical activity levels were at higher risk of developing adverse health conditions, again emphasizing the need for early intervention programs and health education. The research also pointed to the importance of data preprocessing techniques, as handling missing values and variable normalization significantly improved model performance. These findings highlight the strength of data mining techniques in unearthing helpful health patterns and informing evidence-based public health programs that may encourage students to develop healthier practices.

11. Recommendations

In order to make data mining techniques more efficient in prediction and improving student health results, some significant recommendations should be considered. Firstly, improved data quality and preprocessing via the use of advanced imputation techniques for missing data, numeric feature normalization, and feature selection can significantly improve model accuracy. Class imbalance mitigation using oversampling strategies like SMOTE, undersampling, or cost-sensitive learning can improve classification fairness. Cluster performance can be optimized further through hyperparameter optimization, application of ensemble learning techniques such as boosting and stacking, and exploring deep learning models for complex patterns. Silhouette analysis and cluster validation indices must be utilized to provide better interpretability of the clustering results, and domain knowledge can provide insightful information regarding behavioral and health-based segmentation. Further, the translation of results into useful applications via collaborations with public health practitioners can help develop targeted interventions, forecasting dashboards, and monitoring systems that are school-based. Larger-scale research that includes longitudinal studies can provide richer insights into trends in health, and investigating the impact of socioeconomic status can add an important dimension to the comprehension of student well-being. By implementing these recommendations, future research can significantly enhance the precision, applicability, and impact of evidence-based health information.

12. Challengers

Carrying out this research came with various challenges, ranging from data pre-processing to model performance. Among the key challenges was managing missing values, whose removal resulted in loss of data but whose imputation risked the introduction of bias. Class imbalance was also a challenge, particularly for health outcomes variables, and it affected model performance and required resampling. The data set contained numerous categorical features, and these needed transformation through one-hot encoding, hence increasing dimensionality and computational requirements. Data standardization was crucial since measurement scales varied for variables like BMI and frequency of exercise. Methodologically, the best value of k for K-Means was subjective as different values of k yielded different findings, while DBSCAN clustering required parameter tuning of epsilon and minPts to pick out outliers appropriately. Computational complexity was also an issue, especially while training Neural Networks, which required excessive hyperparameter optimization to avoid overfitting. Less complicated models like Naïve Bayes performed poorly for actual prediction, while more complicated models like Random Forest and Neural Networks were harder to interpret. Furthermore, decision trees tended to overfit and required pruning in order to promote generalization. Finally, while clustering methods tastefully

segregated students according to measurements of health, knowledge of cluster interpretations required domain information, thus preventing potential interpretation of raw data patterns as significant health information. Future studies could ease these challenges by applying sophisticated imputation methods, applying synthetic data generation (SMOTE) for class balancing, using hybrid models which combine clustering and classification, and utilizing the explainable AI (XAI) for improving interpretability without compromising high predictive ability.

13.Conclusion

This study efficiently applied classification and clustering techniques in data mining to analyze a data set that spans demographics, diet habits, medical history, exercise frequency, and diet and health beliefs. Decision Tree, Random Forest, Naïve Bayes, and Neural Network classifications identified the most important factors influencing health outcomes, and K-Means and DBSCAN clustering identified patterns within the data. These approaches provided credible information on customer behavior, lifestyle, and potential health issues, validating the importance of data-driven decision-making.

The findings indicate the value of machine learning in identifying trends, segmenting populations, and detecting anomalies to inform public health interventions and policy with a consumer focus. The cluster models also revealed hidden patterns in the data, allowing more targeted strategies within industries such as healthcare, marketing, and social policy formulation.

In conclusion, the research confirms that data mining techniques are useful tools for generating useful information from vast and complex datasets. The results validate our proposed approach by demonstrating its capability to generate useful insights. The study not only adequately answers the research question but also provides opportunities for further research, particularly in the application of sophisticated optimization techniques and incorporating real-time data for even better analysis.